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How to Create AOIs

How to Create Areas of Interest (AOIs) to Analyze Eye Tracking Data in iMotions

- Eye tracking analysis often involves AOI analysis, where areas-of-interest (AOIs) for visual stimuli. Once AOIs are created and defined, you can calculate AOI metrics and create event markers, which indicate when respondents looked at your AOIs.
- AOIs are created and edited in the AOI Window, which is available for any analysis including eye tracking data.
- This article is an abridged version of a more comprehensive article about [Areas of Interest \(AOIs\)](#).

Types of AOIs

In iMotions, you can create Manual AOIs, Dynamic AOIs, and Automatic AOIs

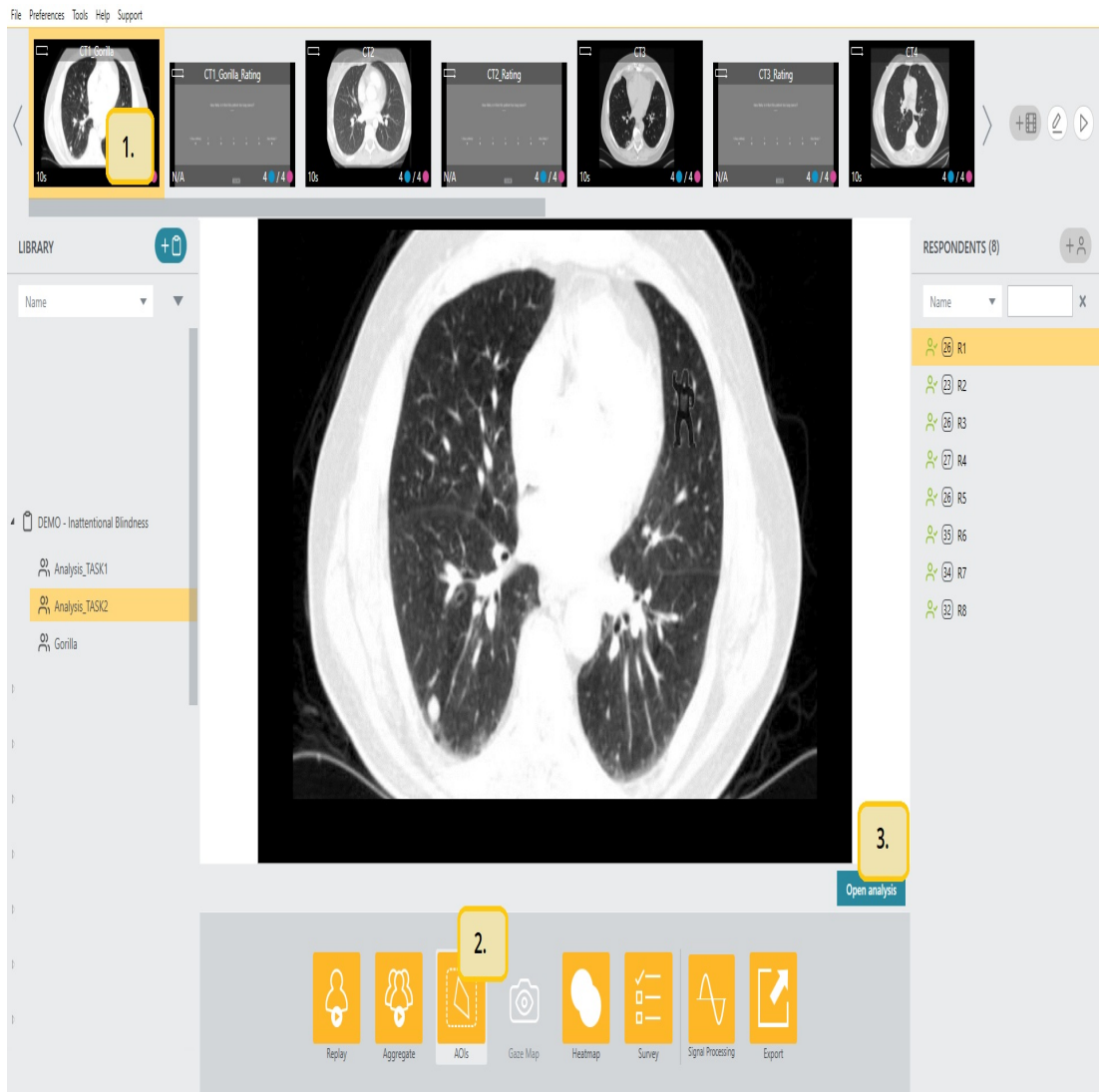
- **Manual AOIs** mark time periods within a *static* stimulus (consistent duration/content across respondents) and are applied to all data recorded with annotated stimuli. Stimulus annotations are generally used to mark relevant time periods within video or image stimuli. Stimulus Annotations must be defined manually.
- **Dynamic AOIs** mark time periods within individual respondent recordings and unique to each respondent. Respondent Annotations can be defined manually or automatically.
- **Automatic AOIs** also mark time periods within individual respondent recordings, but are automatically applied based on event markers in the recordings (available in iMotions 9.2 or newer). Start/End triggers can be selected from any Event Markers available in your study.

AOI Analysis Window

Opening the AOI Analysis Window

1. Once an analysis has been created, select a stimulus/respondent with eye tracking data

2. Click the 'AOI'
3. Click 'Open analysis'



AOI Window Interface

The AOI Analysis Window allows you to create, view, and edit AOIs.

The AOI Analysis Window also allows you to calculate and view AOI Metrics.

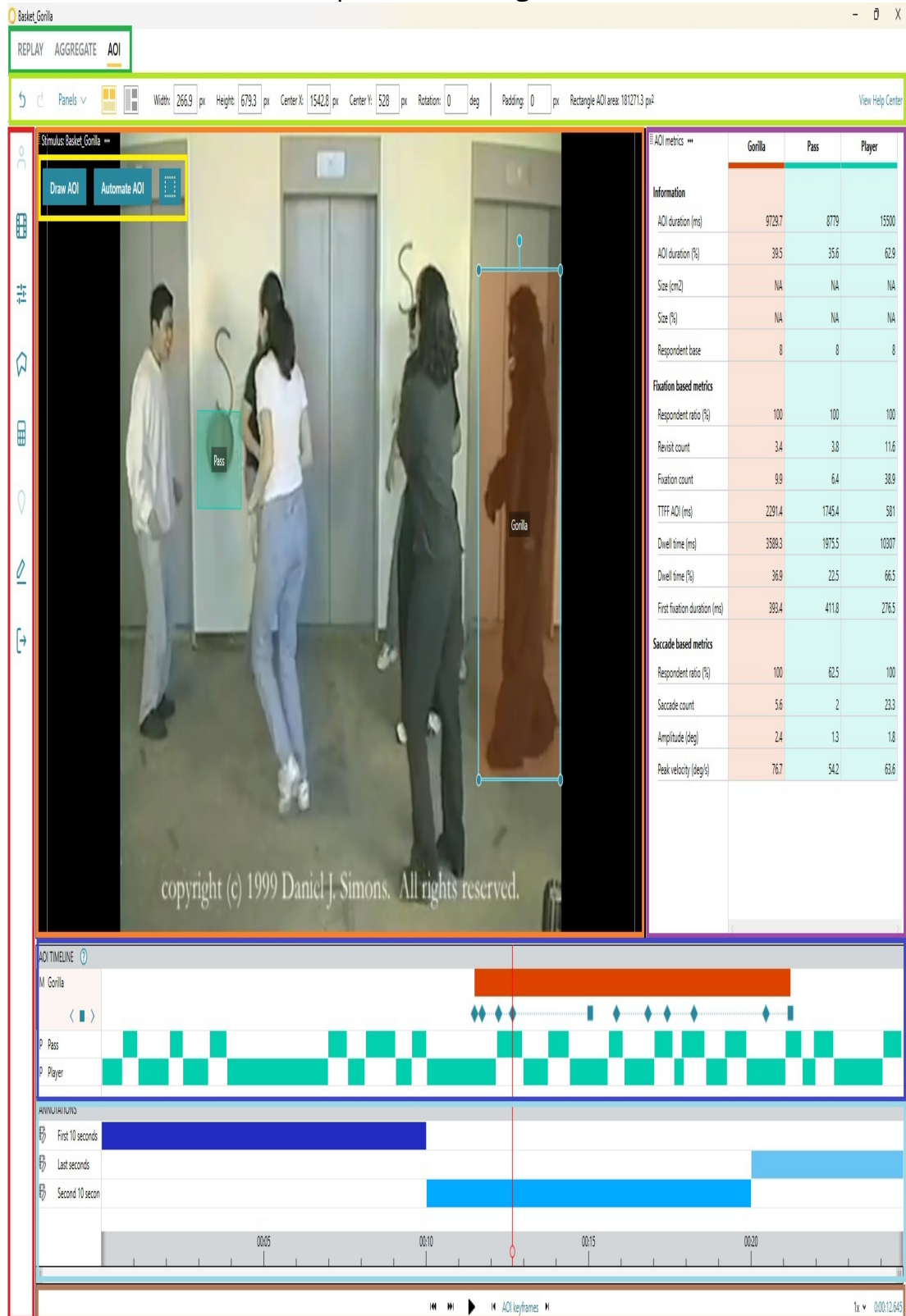
The contents of your Analysis Window can be adjusted with the 'Panels' button or Menu Icons and by dragging panel boundaries.

In this example AOI Analysis Window:

- AOIs are created with the buttons (in the yellow rectangle).

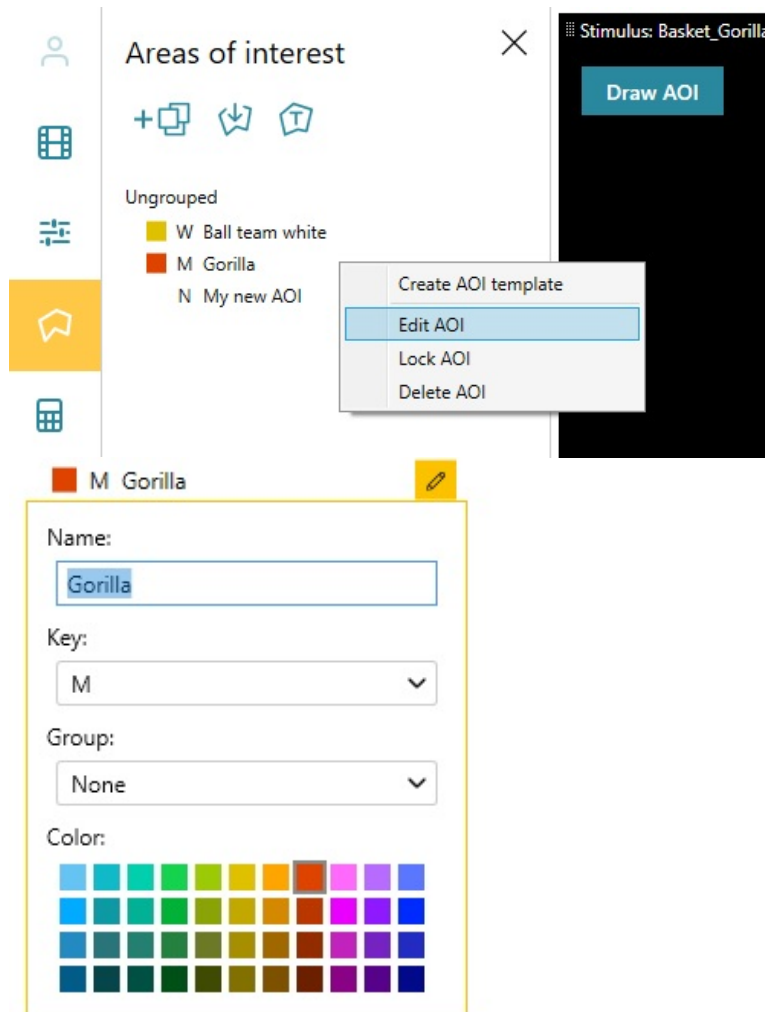
- AOIs appear in the 'Stimulus' panel (in the orange), where they be drawn and edited.
- The AOI Timeline panel (in the purple) shows a list of AOIs and when they are active. The AOI Timeline is used to edit AOI properties and other AOI features (e.g. keyframes for Dynamic AOIs).
- AOI Metrics panel (in the dark purple) shows calculated AOI metrics.
- The Menu Icons (in the red) includes the AOI menu (4th icon) and the AOI Metrics menu (5th icon).
- The AOI Editor (in the green) provides input options for precise definition of your AOIs
- Replay Navigator (in the brown) and Annotations panel (in the blue) are available to conduct AOI analysis.

- You can switch between the 'Replay', 'Aggregate', and 'AOI' interfaces with the buttons at the top (in the dark green)



Areas of Interest Menu

- The 'Areas of Interest' menu provides buttons for creating AOIs and groups. You can also import AOI templates from this menu.
- This menu provides a list of existing AOIs and options for editing, displaying, locking, deleting, and making other changes to AOI properties.
- We recommend allocating a Key to facilitate AOI creation and editing.



Creating Manual AOIs

Manual AOIs are created with the 'Draw AOI' button.

1. Select a drawing 'Tool', which have options for Rectangular, Circular, Polygonal, and Grid AOIs

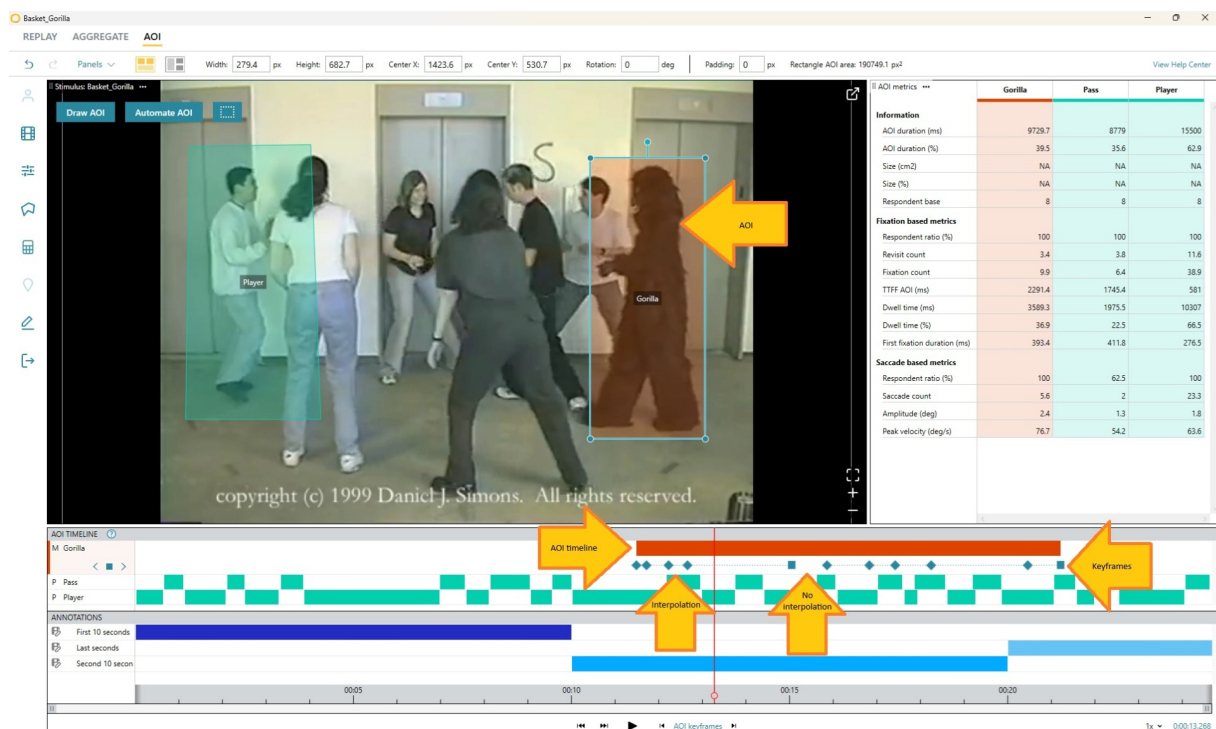
2. Select a color that will be used to display your AOI in the Stimulus Panel and other panels that includes AOI information.



Dynamic AOIs

- Dynamic AOIs are any AOI that moves throughout a dynamic stimulus recording (like a video, a mobile glasses recording, a screen/camera recording, or a website interaction).
- Dynamic AOIs can be made by interpolating between adjustments to Manual AOIs and setting Keyframes for those adjustments. This is typically used to track a moving AOI that has a predictable velocity (e.g. a moving person, vehicles, objects, etc.) or screen/website recordings that contains scrolling. Since objects change location, you can use dynamic AOI to adjust/verify that the AOI remains over the relevant area in the stimulus.
- You can do so by using Automated AOIs with built-in object tracking (available from iMotions 10.1 onward; also requires module for Automated AOIs) or manually editing your AOI so that it is always following the object.
- If you have a completely static stimulus (like an image, survey slide, [Gaze Maps](#), or [website exports](#)), then standard (static/non-dynamic) AOIs are generally sufficient.
- Dynamic AOI terminology:

- **Key frames:** Every time, the geometric shape of the AOI changes (i.e. when it is made bigger), this is marked with a key frame under the timeline. Key frames are always the result of a user action (reshaping, resizing, relocating) or placed by the automation feature, which updates the AOI geometry at a defined sampling rate.
- **Interpolation:** In between key frames, the AOI will either remain at the location it was placed at, or can be move through linear interpolation. The shape of the key frame as well as the dotted line under the timeline tell you whether an AOI is interpolating in between key frames or not.



Manual Workflow for Dynamic AOIs

Dynamic AOI timelines

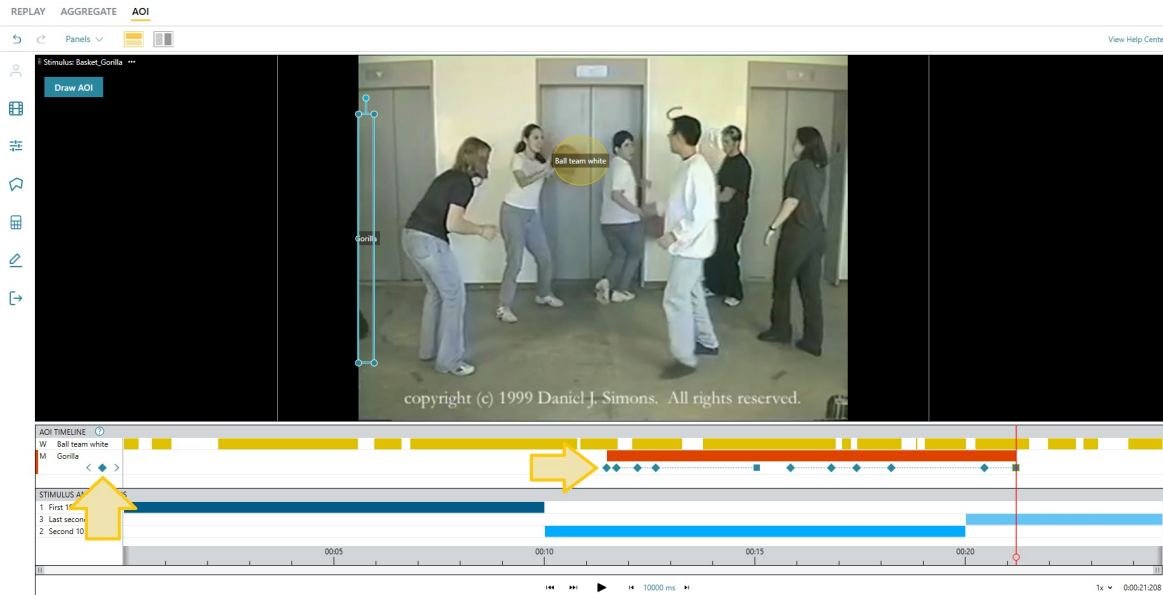
When you define an AOI on a dynamic stimulus, you will notice that a colored line will appear in the 'AOI Timeline' panel. The AOI Timeline indicates when the AOI is active (turned on) or inactive (turned off), which should correspond with the presence or absence of the dynamic objects covered by the AOI.

- Creating an AOI automatically applies the AOI from the selected time point to the end of the recording.
- You can adjust the AOI Timeline into any number of intervals

- Adjusting AOIs with your mouse:
 - Hovering over any AOI timeline will bring up a pen icon , then dragging the interval start/end points.
 - To enter a specific start/end time, right-click for input of specific times
 - To remove or include an interval in your AOI metrics and select 'Activate' or 'Deactivate'.
 - To remove an AOI interval completely, right-click on the interval and select "Delete interval".
- Pressing the keyboard Key (allocated in the AOI menu) creates a start/end point of an AOI interval at the current time point.
- To undo any changes, you can always undo adjustments with the arrow keys or Ctrl+Z on your keyboard.

AOI keyframes

- Every time you edit the AOI (including change in size, shape, or rotation), an AOI keyframe gets placed. This is indicated by a small square or diamond right below the AOI timeline.
- The square indicates an edit at the current keyframe (e.g. change of location), whereas the diamond indicates that the keyframe is also the starting point of a linear interpolation.
- Right next to the AOI name in the AOI timeline panel, you can find a small dialogue to interact with the AOI keyframes 
 - You can jump between AOI keyframes by selecting the option in the Playback Navigator and stepping forward/backward (with the arrow keys or clicking the 'step' buttons)
 - Clicking on the square/diamond will turn interpolation off or on



Turning an Using interpolation

- As a default, the AOI will always remain in its current state (i.e. at its current location with the current size) until you edit it. However, especially if you want to track a movement across many frames, you can use linear interpolation to make the AOI move from one location to the other without having to manually adjust it on every frame in between.
- You already read about AOI keyframes in the previous section. If you edit a dynamic AOI, e.g. change its location, this animation event will be indicated with a little square below the timeline. The AOI will remain at its original location and then suddenly appear at the new location at the indicated keyframe.
- If you want your AOI to move from one location to the other (e.g. because you want to track a linear motion), you can activate interpolation. You can either do so by
 - clicking on the diamond icon in the dialogue next to the AOI name < ◆ >
 - or by right-clicking on the AOI shape and selecting "Interpolate"
- If interpolation is turned on at one keyframe, all subsequently placed keyframes will also activate the interpolation. If you want to stop this and do not want your AOI to automatically be moved, you can turn interpolation off by
 - clicking on the square icon in the dialogue next to the AOI name < ■ >

- or by right-clicking on the AOI shape and ticking off "Interpolate"

Automated AOIs

From iMotions 10.1 and onward, it is possible to place AOIs on dynamic stimuli automatically using computer vision.

- This feature requires the activation of the Automated AOIs module and works best on PCs with an [NVIDIA GPU of compute capacity between 6.1 and 12.x](#).
- Note: Prior to initial use, iMotions will prompt the user to install the Automated AOIs package - this occurs the first time you click on 'Automated AOI'.

Defining and Processing Automated AOIs

Working with **Automated AOIs requires the definition of at least one *input frame***, i.e. a frame in the video where the object that you wish to track is clearly visible.

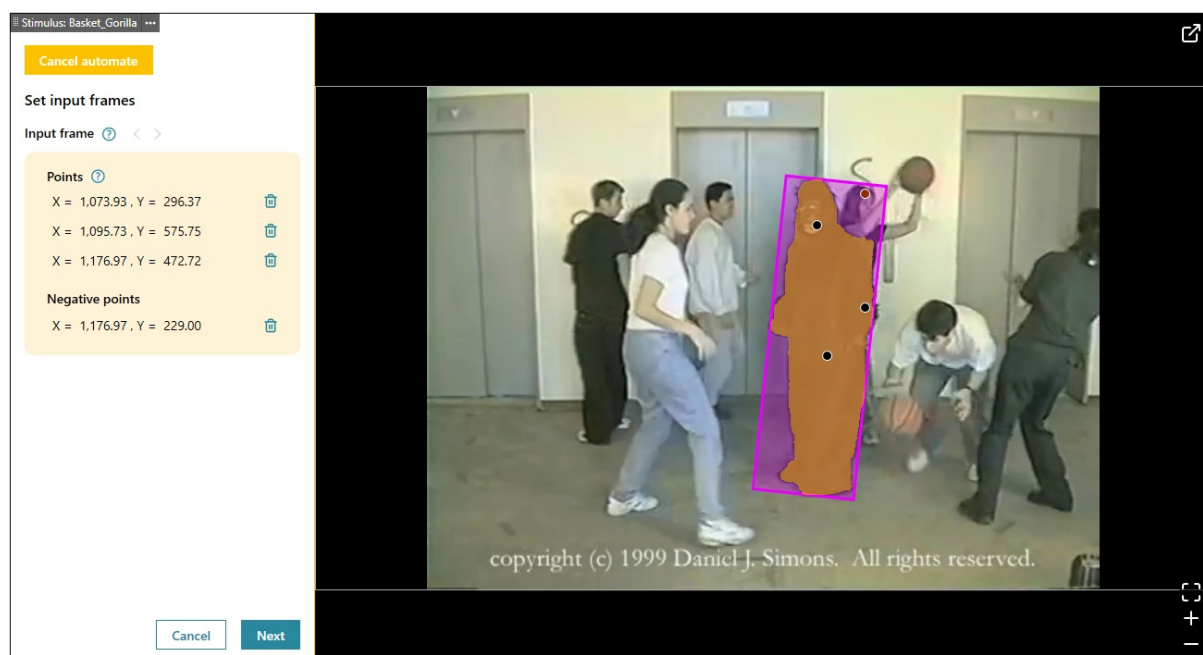
You can choose to select more input frames if your object is changing a lot throughout the recording.

To define an AOI through the automation feature, you have five options:

1. Right click on the object visible in the stimulus and select "Automate"
2. Click on "Automate AOI" > New AOI
3. If you already have started drawing an AOI manually (rectangle only), right click on the AOI and select "Automate"
4. If you already have an existing AOI, or defined an AOI on other respondents, click on "Automate AOI" > Existing AOI
5. If you already defined an AOI on other respondents, click "Automate" on the timeline

On the input frame, you select the object by clicking on it until it is entirely covered by an orange mask. Black dots indicate where you clicked. If the mask exceeds the object, you can *Ctrl+click* to set negative points (red) to tell iMotions that this is not a part of the object.

The rectangle around the mask is a preview of the resulting AOI. Please note that you need to process the video in order to get the actual AOI over the entire video.



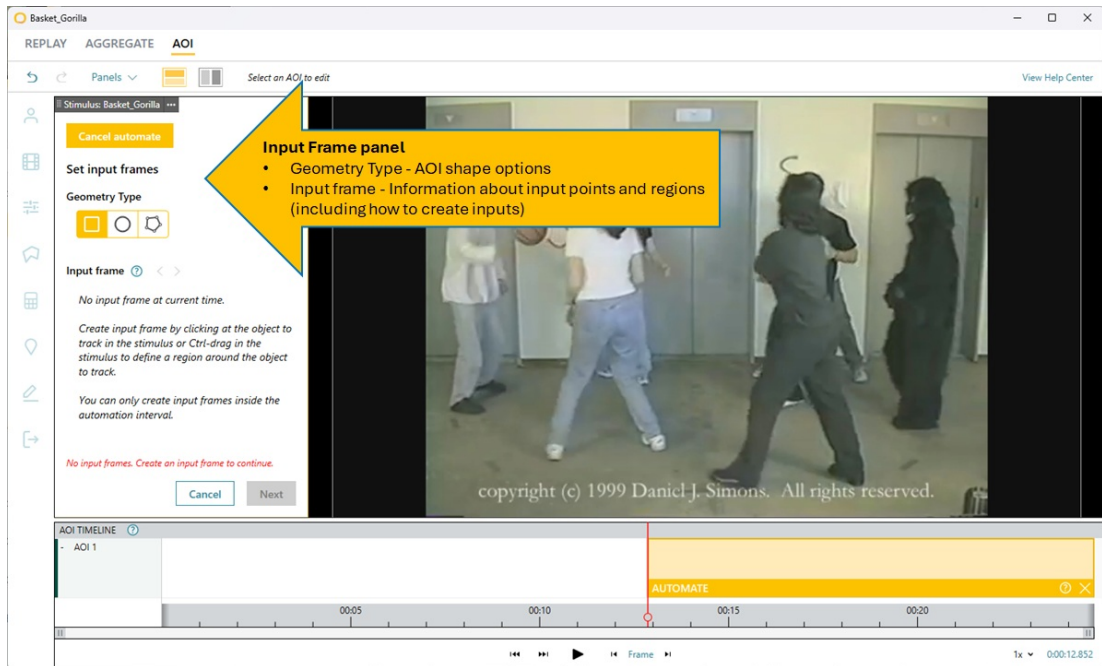
Creating Input Frames

Input Frames are defined by input points or regions (in iMotions 10.1.11 or newer; previous versions only have the point option).

Automated AOI shape options include **Rectangles, Ellipses, and Polygons** (in iMotions 10.1.8 or newer; previous versions only have the rectangle option).

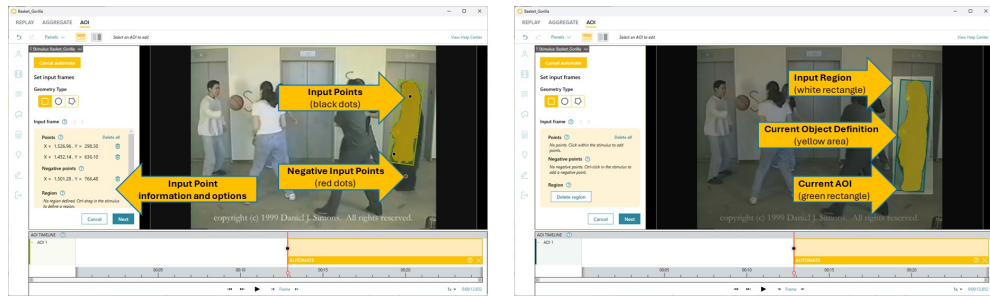
The **Input Frame panel** can be found in the AOI Analysis Window and:

- Allows definition of Input Frames and provides information about user interface
- Allows selection of Automated AOI shape (Geometry Type)
- Displays information about existing input frames, points, and regions
- Allows adjustment of Automated AOI Timeline range



Input Frame panel

1. **Open the Input Frame panel** (steps from AOI Replay Window):
 - a. Navigate to a video frame that includes the object that will define the Automated AOI.
 - b. Right-click on the object and select 'Automate new AOI' (to open the panel for a new AOI and initial input point).
 - c. Alternatively, click Automate AOI and select 'New' (to open the panel for a new AOI without creating an input point).
 - d. Existing input frames are indicated with a black dot in the AOI Timeline.
2. Select **Geometry Type**: Rectangular, Elliptical, or Polygonal (options depend on iMotions version, as noted above)
3. Add **Input Points or Regions**:
 - a. **Left-click** - Adds '**positive points**' that are associated with the AOI object
 - b. **Right-click** - Adds '**negative points**' that are *not* associated with the AOI object
 - c. **Ctrl+Left-click and drag** - **Defines region** associated with the AOI object



Input Frames with Input Points (left) and Input Region (right)



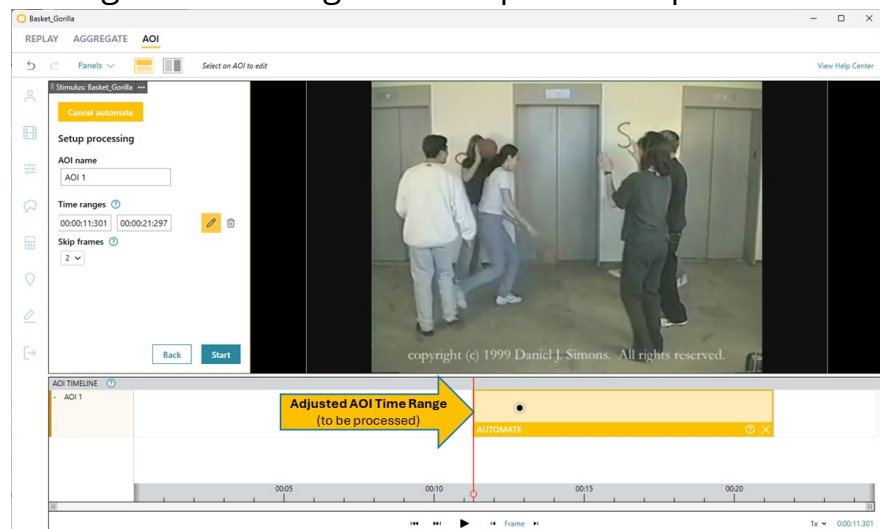
Geometry Type options

4. Adjust **Skip frames** (as needed):

- Skip frames = the number of frames between Automated AOI key frames, processing, and definitions.
- The AOI will be interpolated between the resulting key frames.
 - The default is generally recommended (i.e. 2 skip frames).
 - Lower Skip frames (processing more frames) are needed if the AOI object moves constantly or quickly.
 - Higher Skip frames (processing fewer frames) can be used if the AOI object remains in a stable, predictable location or moves slowly.
 - Increasing Skip frames can help reduce processing time (by skipping X frames and reducing the number of processed frames).
 - Increasing Skip frames provides 'coarser' (but faster) processing and would interfere with object tracking for movements that occur between processed frames (especially non-linear movements).

5. Adjust **Automated AOI time range**:

- a. Include all times where the AOI object should be detected (e.g. when the object is expected to be visible) - only the highlighted times will be processed.
- b. Exclude this times that are not relevant for your AOI analysis.
- c. Exclude times when the object is not expected or out of view (optional but recommended, to reduce processing times).
- d. **Options for setting Automated AOI time range:**
 - Clicking-and-dragging the time range in the AOI Timeline panel
 - Right-clicking to activate and deactivate
 - Editing the time range in the Input Frame panel



6. Adjust **Skip frames** (as needed):

- Skip frames = the number of frames between Automated AOI key frames, processing, and definitions.
- The AOI will be interpolated between the resulting key frames.
 - The default is generally recommended (i.e. 2 skip frames).
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 - Increasing Skip frames provides 'coarser' (but faster) processing and would interfere with object tracking for movements that occur between processed frames (especially non-linear movements).



Reviewing and correcting Automated AOIs

Once the processing is completed, iMotions will return an AOI geometry per processed video frame, indicated by the AOI key frames on the timeline. Unless you chose to process all frames, i.e. to skip 0 frames ahead, iMotions interpolates the AOI geometry in between key frames.



We strongly recommend you to replay the entire processed time range and visually inspect whether the AOI got placed in a satisfactory way.

You can easily step from keyframe to keyframe with the arrow buttons or arrow keys on your keyboard (requires you have an AOI selected).

Please note that upon completion of the processing, iMotions will default to stepping per AOI keyframe.

Once you have reviewed the AOI, we recommend you remove the "Automation Completed" message by ticking it off.

If you are not satisfied with the resulting AOI, you have two options:

- manually editing the AOI on the keyframes
- re-running selected time ranges with new input frames

To manually correct an AOI, find the key frames where you want to change the AOI and edit it through resizing, rotating, relocating, or turning it off or on.

These are the same steps as drawing an AOI manually (see above) and your changes will overwrite the result of the automated processing.

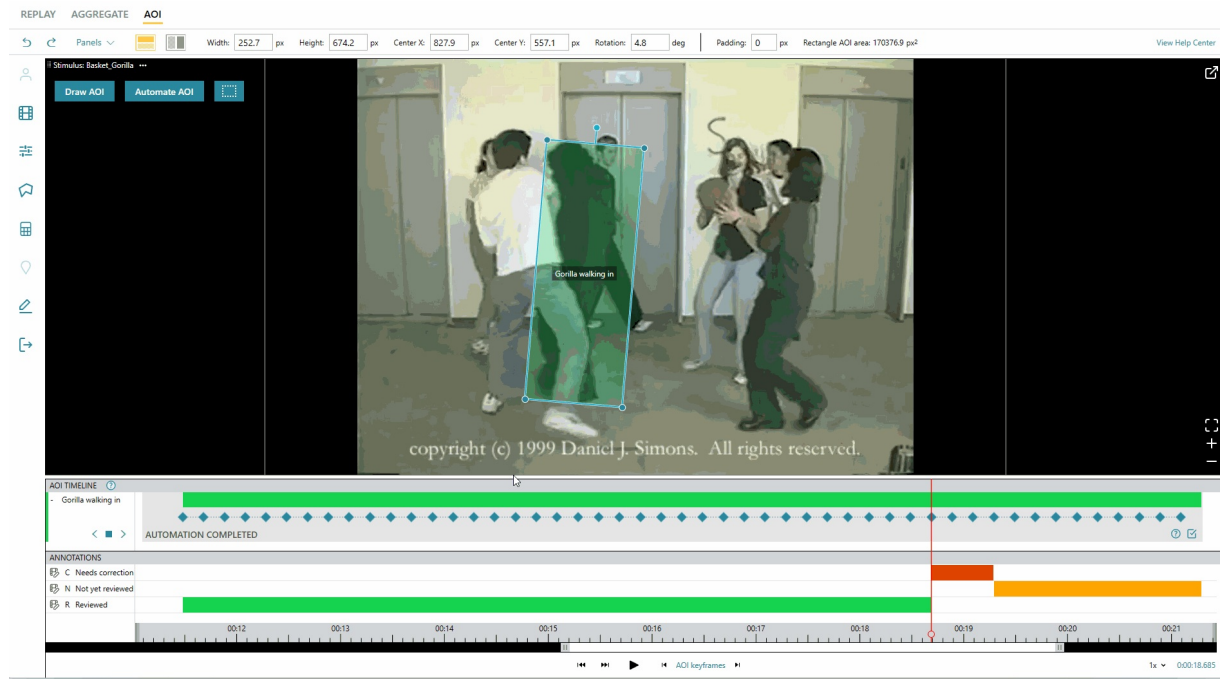
It is also possible to completely clear the AOI timeline as well as key frames in a selected time range by *right clicking on the timeline > Select range to delete..*

If you wish to re-run the automated AOI processing on selected time ranges, this will allow you to provide new input frames and select new time ranges.

The old results in these time ranges (both key frames from the automation as well as manual changes) will be overwritten and replaced with new results. This action can be undone by using the arrow buttons or hitting *Ctrl+Z* on your keyboard.

To reprocess time ranges of an already existing AOI, you can

- click on "Automate AOI" > Existing AOI and then select the AOI
- right click on the AOI and select "Automate"



Tip

When new input frames and time ranges to an AOI that already has an AOI drawn (e.g. from previous processing), then this existing AOI will be shown in black while the preview of the new AOI (the result of the input frames you set) will be shown in color. You can choose to show or hide the existing AOI geometry by right clicking on it.



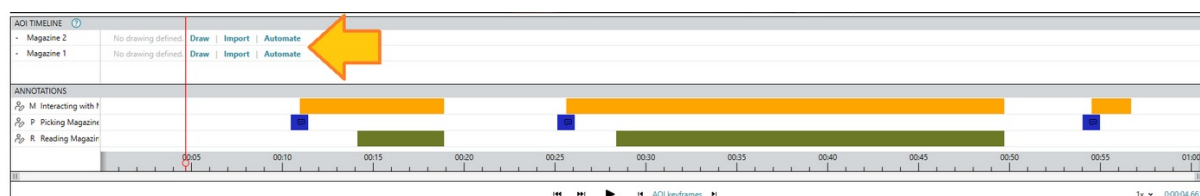
Applying AOIs Across Respondent Recordings

If you are working with recordings that are unique for each Respondent, then you will have to define AOIs individually for every single respondent.. This applies to the following study stimuli:

- Recordings from mobile eye-tracking glasses
- Recordings from a scene camera setup, including setups using a frame grabber
- Screen recordings
- Website recordings
- VR stimuli

Existing AOIs are applied to all respondents to be defined per respondent with the following options in the AOI Timeline:

- Drawing AOIs per respondent
- Importing an already existing AOI
 - Reusing a shape that you already defined for another respondent. This will apply the initial shape and import the timeline. Note: Dynamic adjustments are not included and must be applied, if they differ from the initial/imported AOI and its timeline.
 - Importing an AOI template
- Start a new automation for this respondent



Importing AOIs from CSV file

AOI definitions can be imported into your study

- *Respondent Name*
 - Allowed values are: Text input (or blank)
 - The column can only be left blank if it constitutes a keyframe succeeding an initial dynamic AOI definition row

- For static AOI definitions, the name should correspond to one valid respondent across the file (as a static AOI will be applied for all respondents) compared to a dynamic AOI which is unique for each respondent
 - Is case sensitive
- *Stimulus Name*
 - Allowed values are: Text input (or blank)
 - The column can only be left blank if it constitutes a keyframe succeeding an initial dynamic AOI definition row
 - Is case sensitive
- *AOI Name*
 - Allowed values are: Text input (or blank)
 - The column can only be left blank if it constitutes a keyframe succeeding an initial dynamic AOI definition row
- *Color*
 - Allowed values are: Text input or blank
 - Optional
 - Must be specified in hex code
- *Group*
 - Allowed values are: Text input or blank
 - Optional
 - Includes the AOI within the named group
- *Timestamp (ms)*
 - Allowed values are: Text input
 - Allowed values are integers or floats (numeric values with decimals indicated by a ".").
 - Start time of the AOI relative to stimulus start (Video or Screen Recording), in milliseconds
 - For a static AOI definition, the timestamp is ignored (but still needs to be set, e.g. to "0")
- *Is active*
 - Allowed values are: Text input (dynamic AOI) or blank (static AOI)
 - A value of "1" will activate the AOI
 - A value of "0" will de-active the AOI
- *Points*
 - Allowed values are a series of value pairs, X and Y coordinates - either as integers or floats (numeric values with decimals indicated by a ".").
 - X and Y values are separated by ";" and pairs using ";;"

- Coordinates that make up an AOI shape (polygon)
- Minimum of 3 points must be defined
- *Translation*
 - Allowed values are value pairs, X and Y offsets - either as integers or floats (numeric values with decimals indicated by a ".")
 - X and Y values are separated by ";"
 - Translation effectively means offsetting the shape with the provided coordinate values
- *Scale*
 - Allowed values are value pairs, width and height (AOI) scaling factors - either as integers or floats (numeric values with decimals indicated by a ".")
 - Width and height values are separated by ";"
- *Rotation*
 - Allowed values are: Text input
 - Values are provided in Degrees - either as integers or floats (numeric values with decimals indicated by a ".")
- *Interpolate*
 - Allowed values are: Text input (dynamic AOI) or blank (static AOI)
 - A value of "1" will enable interpolation to the next key-frame
 - A value of "0" will disable interpolation to the next key-frame

Download sample [static AOI](#) or [dynamic AOI](#) file.

AOI Metrics

- The AOI Metrics Menu allows initial calculation of AOI metrics and a list of available metrics. Information about each metric is available by hovering over a metric, then the "?" icon that appears.

- Clicking the icon to the left of each AOI metric will control whether it appears in the AOI Metrics Panel.

	Left lung	Right lung
Information		
AOI duration (ms)	9997.1	9997.1
AOI duration (%)	100	100
Size (cm2)	88.9	88.9
Size (%)	16.5	16.5
Respondent base	5	5
Fixation based metrics		
Respondent ratio (%)	100	100
Revisit count	4.2	4
Fixation count	9.2	16.2
TTFF AOI (ms)	2.3	244.1
Dwell time (ms)	3527.9	5921.6
Dwell time (%)	35.3	59.2
First fixation duration (ms)	597.9	589.6
Saccade based metrics		
Respondent ratio (%)	100	100
Saccade count	4.2	13.4
Amplitude (deg)	2.9	2.4
Peak velocity (deg/s)	76.9	71.9

Calculate and View AOI Metrics

- The 'AOI Metrics' menu initially contains a 'Calculate' button. Clicking 'Calculate' will begin processing metrics for the AOI in your analysis.
- If AOI metrics have been calculated, a list of available metrics is displayed, along with display settings and information about each metric.

Export AOI Metrics

- AOI metrics can be copied directly from the 'AOI metrics' panel, by clicking the '...' icon then 'Copy table as CSV'.
- AOI Metrics can be exported in Sensor Data and Analysis CSV files
- To include AOI analysis in Sensor Data:
 - Use the Sensor Data Export window and enable File Info options for AOIs.
 - These options include indicators for AOI dwells in the resulting CSV file.
- The Analysis Export window includes a section for AOI Metrics. This is used to export AOI metrics CSV files for all AOIs, intervals, stimuli, and respondents in the analysis. This creates two CSV files.
 - Aggregate metrics
 - Respondent metrics

AOIs

Select all | Select none

☒ Include AOI markers

☐ Show separate AOI columns per Stimulus

☐ Combine AOI markers

☐ Show AOI markers as text

☐ Include AOI comments & instances

BUILT IN PROCESSING | R PROCESSING

Export individual files

Individual metrics export | Single multi-modal export

AOI

AOI Metrics Export

SETTINGS

Please select export type(s):

Select all | Select none

☒ Individual AOI metrics

☒ Aggregated AOI metrics

☒ Fixation & saccade tables

☐ Transition matrices

Threshold for dwell classification (ms) 100

Amount of first fixations to exclude 0

☐ Exclude fixations starting before AOI onset or ending after AOI offset

Annotations:

Select all | Select none

☐ Include annotations in AOI metrics

☐ Include annotations in fixation & saccade tables

☐ Create transition matrices based on annotations

Scenes:

Select all | Select none

☐ Include scenes in fixation & saccade tables

DESCRIPTION

Produces files export for AOI metrics.

More information about this notebook and its parameters can be found in the Help Center